

Brainstorming 3rd Study

Sustainable and Scalability 4P (Problem Finding, Problem Exploring, Problem Generating and Problem Solving) **with AI-Driven Automated Exploratory Inquiry for Physics Learning in Authentic Contexts**

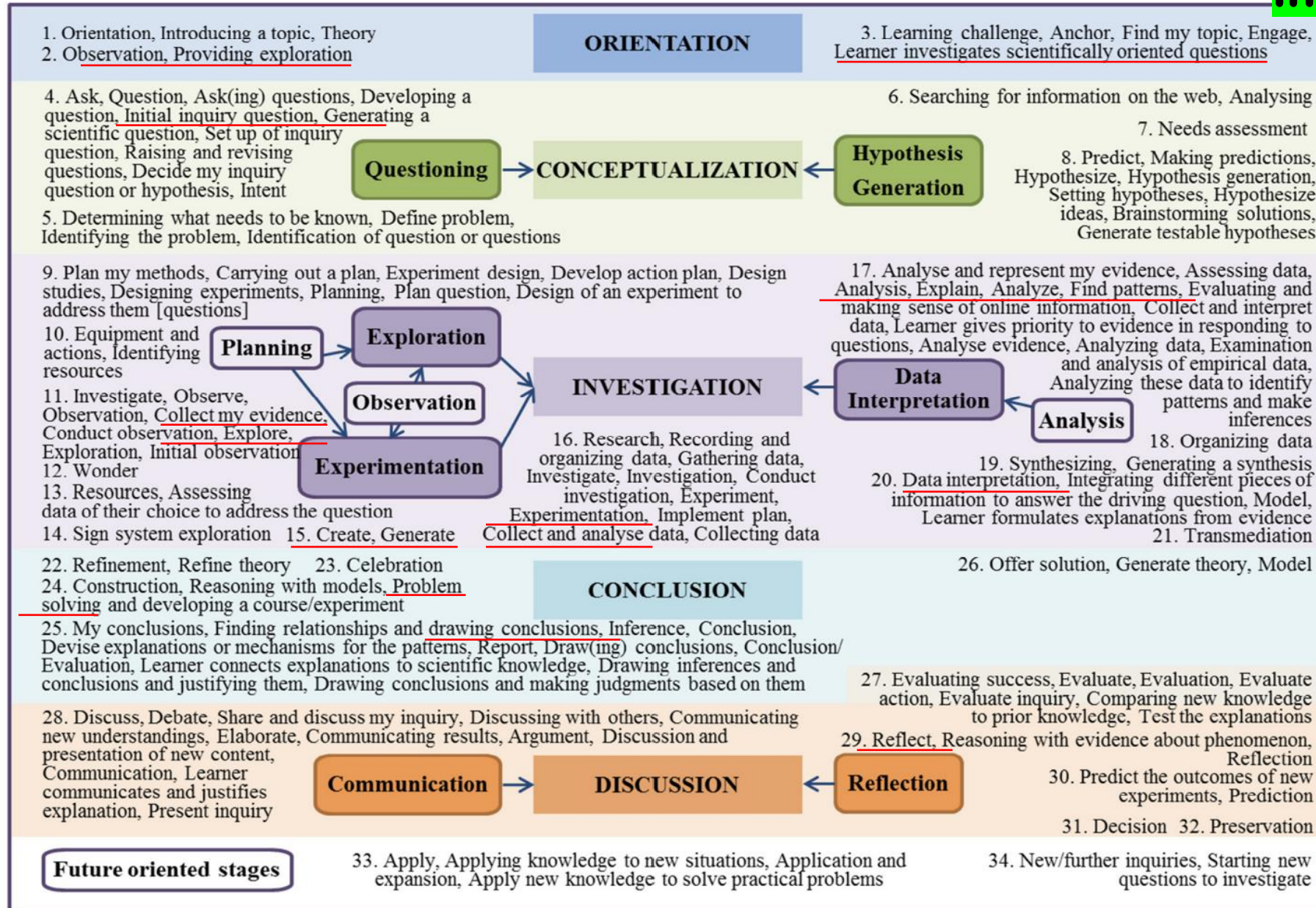
P-MAGIC V.2

By : Vinsa

25/12/2025

- **Problem-Based Learning (PBL)**
- **Exploratory learning (EBL)**
- **Inquiry-Based Learning (IBL)**

inquiry-based learning (IBL)



Highlighting Student Activities in P-MAGIC V2

11 Highlights

Pedaste, M., Mäeots, M., Siiman, L. A., de Jong, T., van Riesen, S. A. N., Kamp, E. T., Manoli, C. C., Zacharia, Z. C., & Tsourlidaki, E. (2015). Phases of inquiry-based learning: Definitions and the inquiry cycle. *Educational Research Review*, 14, 47–61.
<https://doi.org/10.1016/j.edurev.2015.02.003>

Fig. 2. Visualization of the process by which the initial 34 groups of inquiry activities were merged into bigger groups and finally into five general inquiry phases.

problem-based learning (PBL)

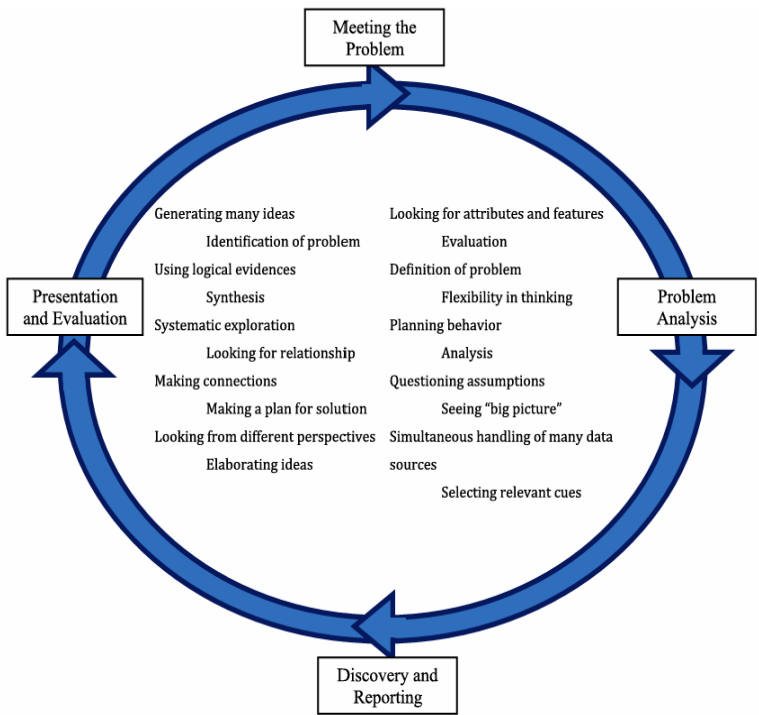
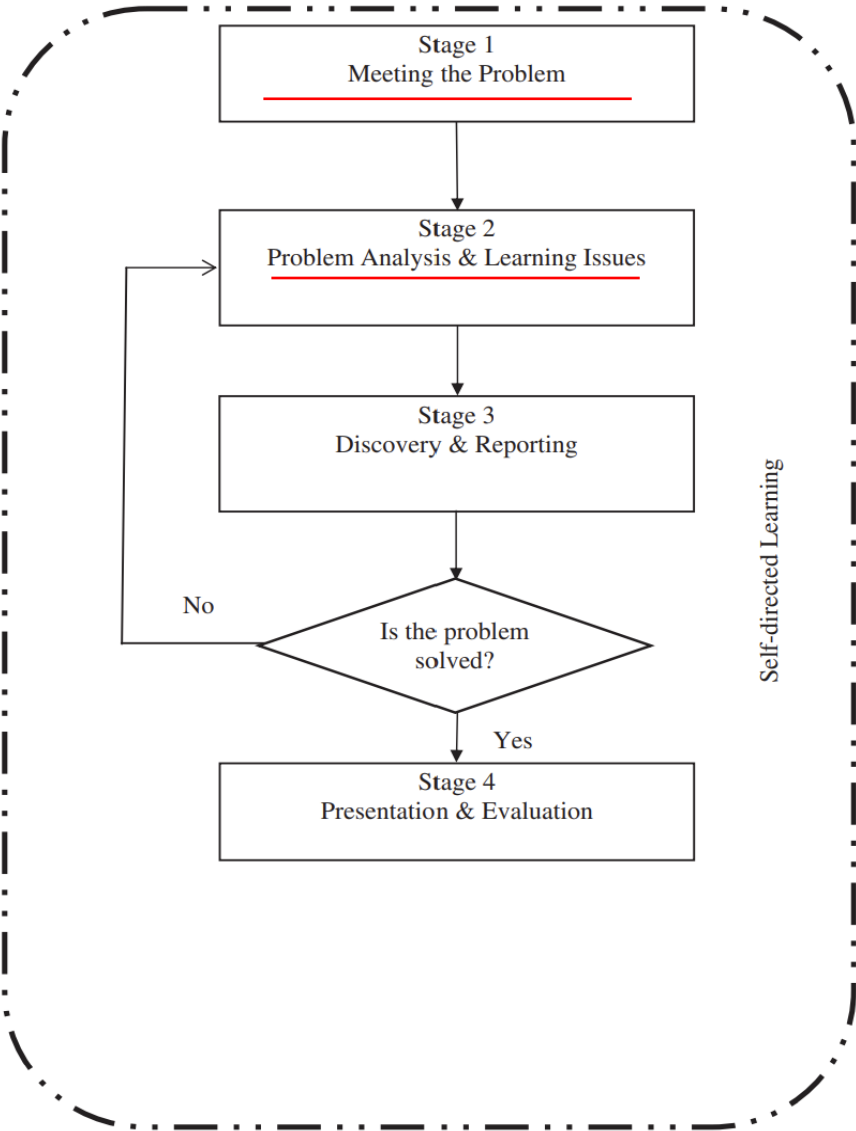


Figure 2. Conceptual framework for identifying cognitive functions in the PBL stages.

The four stages of PBL are: **(1) Meeting the problem**—learners are introduced to a scenario, identify key facts, and propose initial ideas. **(2) Problem analysis and learning issues**—learners connect prior knowledge to the scenario, generate questions and hypotheses, formulate a group problem statement, and divide learning tasks. **(3) Discovery and reporting**—members research their assigned tasks, share findings with the group, and refine understanding through peer teaching, questioning, and critique. **(4) Solution presentation and evaluation**—the group presents the problem statement, hypotheses, and solutions to the tutor and peers, justifies their reasoning, responds to questions, and reflects on their learning to support higher-order thinking and self-regulated learning (Garrison, 1993; Masui & De Corte, 2005).

Highlighting Student Activities in P-MAGIC V2

2 Highlights

Figure 1. The 4-stage PBL cycle. Adapted from Tan (2003).

Exploratory-based learning (EBL)

- **extends Kolb's experiential learning model:** Kolb (1984) describes learning as a cycle, but it is largely descriptive and grounded in “real-world” lived experience.
- **Exploratory learning model adds “Exploration” as a core learning construct:** the cycle becomes five steps—**Experience → Exploration → Reflection → Abstract Concept Formation → Testing/Reinforcement**

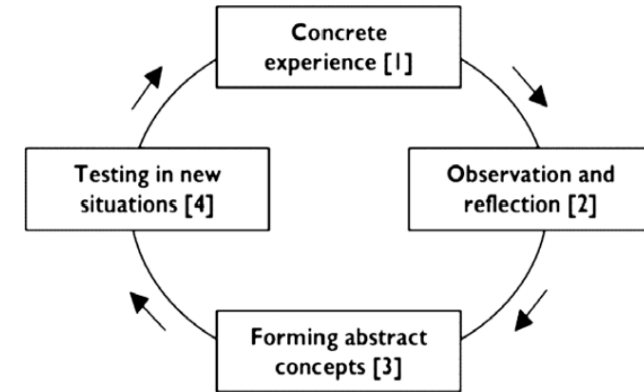


Fig. 1. Kolb's experiential learning cycle (Source: Kolb, 1984).

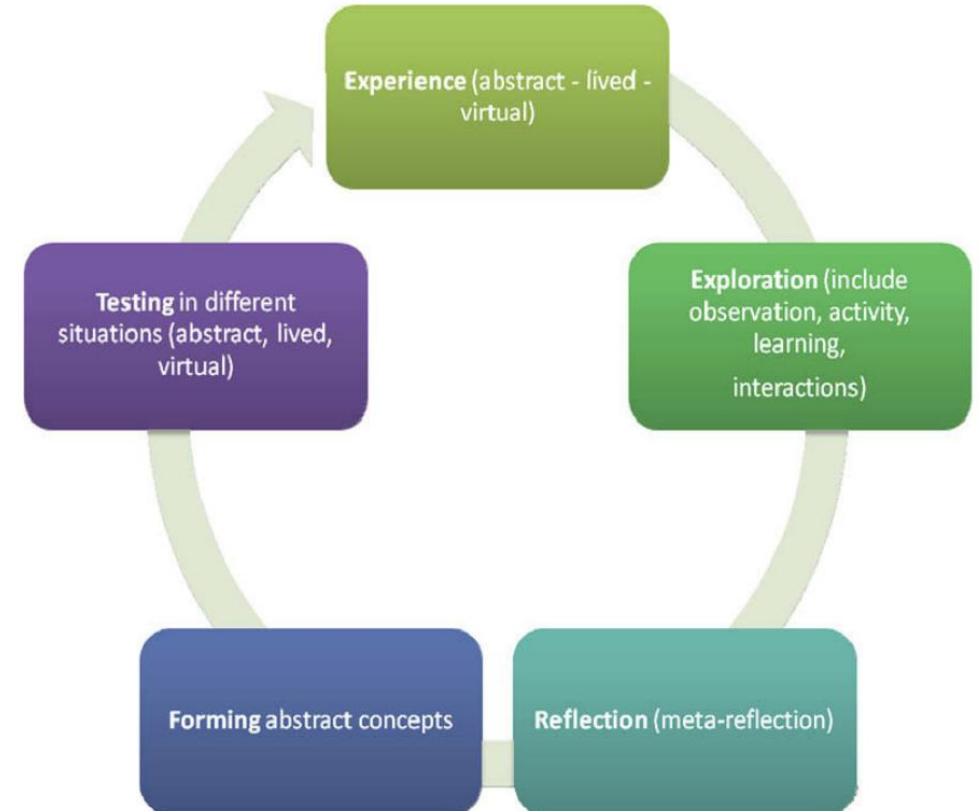


Fig. 2. The exploratory learning model.

Exploratory-based learning

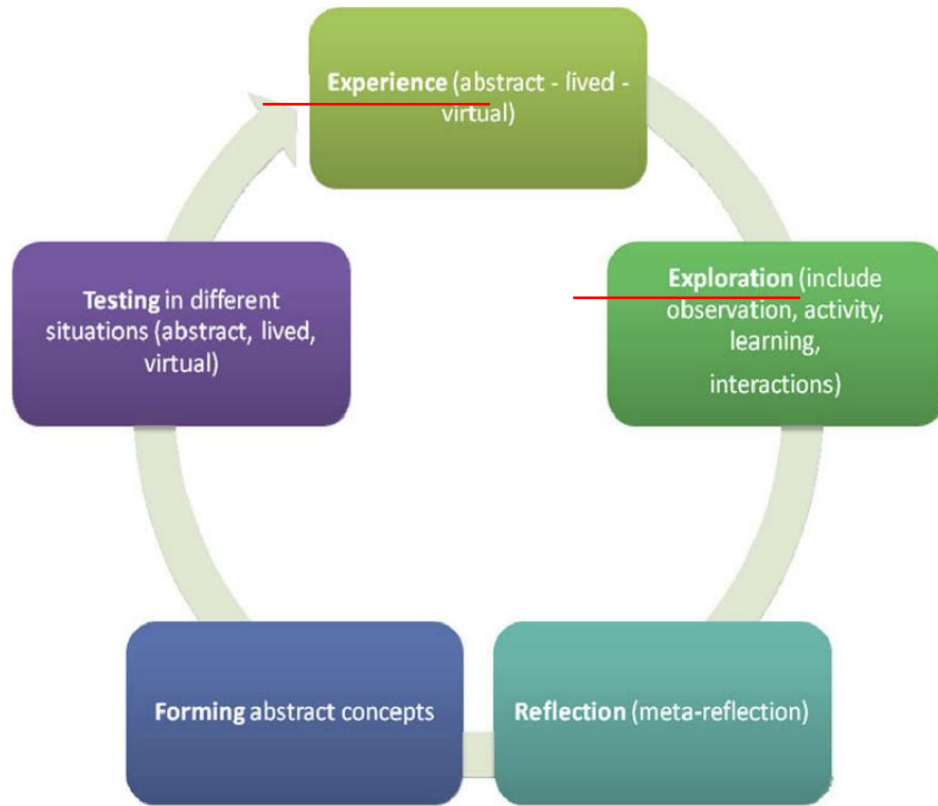


Fig. 2. The exploratory learning model.

- **Experience**

This is the first step, when learners face an event that gives them something to learn from. It can be a real activity (like doing a lab or observing their surroundings) or a virtual activity (like playing a simulation/game or doing VR role-play). The main point is that learners do something and get results, feelings, and social interactions that become learning material.

- **Exploration**

In this step, learners try things out to see what happens. They actions, collect information, ask questions, work with others. The goal is not to find the “correct answer” yet, but to understand more by trying and interacting.

- **Reflection**

Here, learners pause and think about what happened. They compare what they expected with what really happened, and they think about why something worked or did not work.

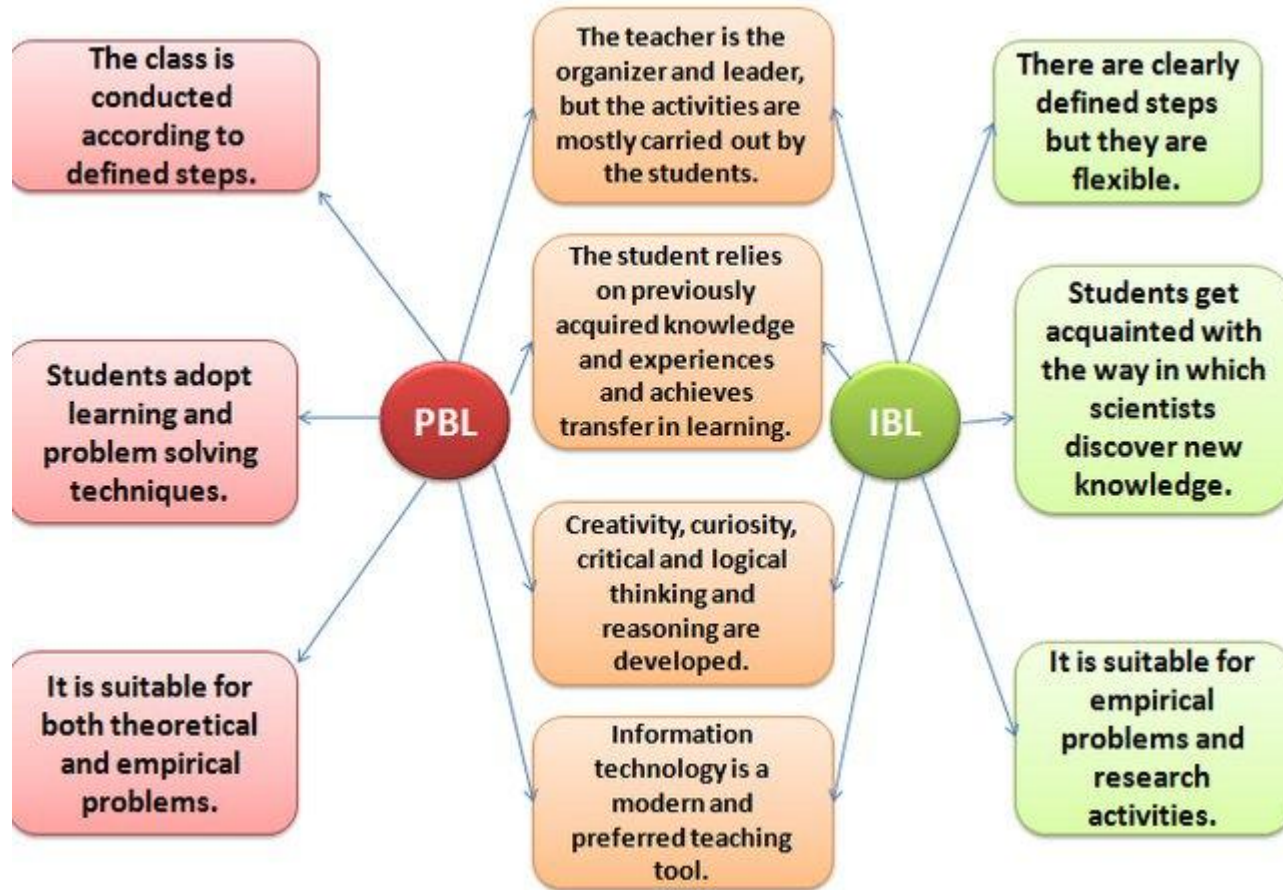
- **Forming abstract concepts**

In this step, learners turn their reflection into general ideas. They build principles, strategies, or simple rules that explain what the experience means. Examples, diagrams, guiding questions, or theory can help learners make these ideas clearer.

- **Testing (experimentation or reinforcement)**

Finally, learners use their new ideas in another situation to see if they still work. They might try again with a new strategy, apply it to a real task, or practice it several times to remember better. The goal is to confirm the idea, improve it, and make it useful in different situations.

Similarities and differences between **problem-based** and **inquiry-based learning**



Problem-based learning (PBL) and inquiry-based learning (IBL) are very similar. Both help students learn well by being active and solving real tasks.

The main differences:

- Steps:** IBL is more flexible and can repeat in a cycle (ask → explore → learn → ask again). PBL usually follows **clear, fixed phases** in the lesson.
- Type of task:** PBL can use many kinds of problems (theory or practical). IBL more often uses **research, experiments, and investigation**.
- Teacher role:** In both, the teacher plans the lesson and guides students, especially at the beginning. Over time, students become more independent, and the teacher mainly monitors and gives feedback.
- Student outcome:** PBL helps students practice problem-solving, learning skills, and teamwork. IBL also builds those skills, but adds stronger understanding of **scientific thinking** and encourages curiosity as a habit.

Problem-Based Learning (PBL)

The **driver** is a **problem** that must be addressed. Learning is organized around producing a **solution, decision, or deliverable** (proposal, diagnosis, design, plan). **The problem usually comes first** and forces students to identify what they need to learn (“learning issues”). **Exploration happens**, but it is typically **goal-directed** toward solving the problem.

Exploratory learning (EBL)

The **driver** is *exploration*—learners learn by navigating a space (real or virtual), experimenting, and discovering patterns. The starting point can be **any experience**, not necessarily a defined “problem” like in the PBL. The outcomes can be **strategies, tacit skills, or behavioral patterns**, not only a “solution.”

Inquiry-Based Learning (IBL)

The **driver** is a **question (or curiosity)**. Learners investigate like researchers: ask questions, explore, collect data, analyze evidence, solving the problems, and build explanations. **Exploration exists**, but it is usually tied to **evidence generation** (observations, experiments, sources).

ADD OUR HIGHLIGHT: CONNECT TO AUTHENTIC CONTEXT, our unique, their surroundings, can motivate them, stimulate curiosity much more than IBL, advance exploration in their daily life.

Combine together IBL and EBL. Authentic exploratory, we give particular topic (known problem)

Synthesis them together. **Sustainability SDG meet one important keyword. (exploratory) sustain their motivation, scalable : many many scenarios, many places. Exploratory inquiry based question finding in authentic context can meet scalable and sustainable.**

Based on the 3 theoretical frameworks above, our approach aligns more with inquiry-based learning

Framework

This demonstrates our unique contributions to achieving sustainability and scalability through problem exploration, finding, generation, and solving in authentic contexts.

1. Problem Finding

Students are encouraged to **find physics concepts** in their surrounding environment related to centripetal acceleration. Using P-MAGIC, they can observe centripetal principles. This leads to **increased active engagement** and **motivation** to continue learning. This phase leverages **P-MAGIC models** to detect **misconceptions** and assist students in finding relevant and interesting physics problems.

2. Problem Exploring (exploratory) find value. Exploratory Inquiry learning

In this stage, students are encouraged to **explore their choice of problems in problem finding stage**. Using P-MAGIC, they can experience and analyze the real world data centripetal principles through hands-on experiments. This leads to **increased active engagement** and **motivation** to continue learning.

3. Problem Generating

Next, **P-MAGIC generates problems** related to the object already find and explored. **Multimodal problems** including annotation images, text, graphs, or table, all aimed at testing a deeper understanding of the concepts. Include : **Problem difficulty level**.

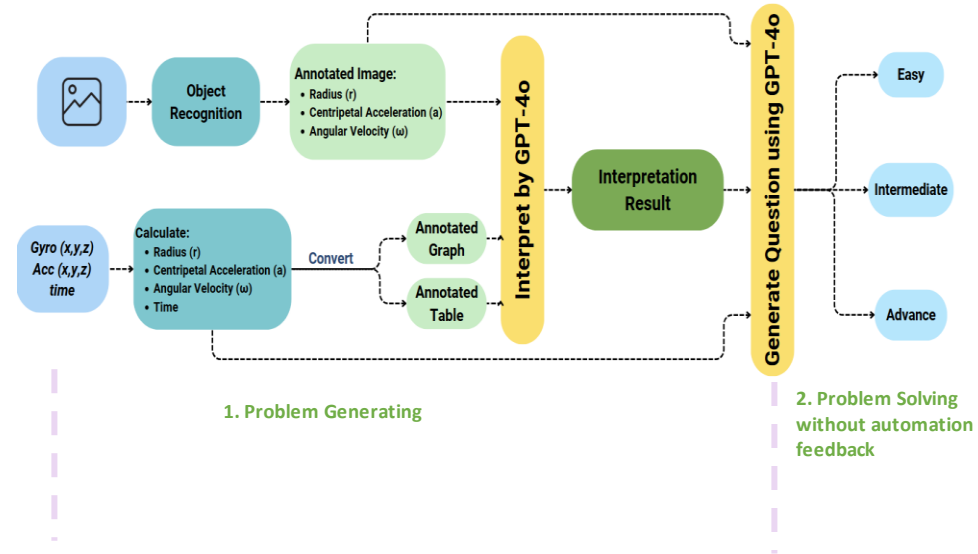
4. Problem Solving

In this phase, students **solve the problems** generated by the system, receiving adaptive feedback. The **adaptive feedback system** offers clarifications and motivation when students encounter difficulties or show declining engagement. **Solutions based on real-world data** ensure **sustained learning**, connecting physics theory to real-world applications.

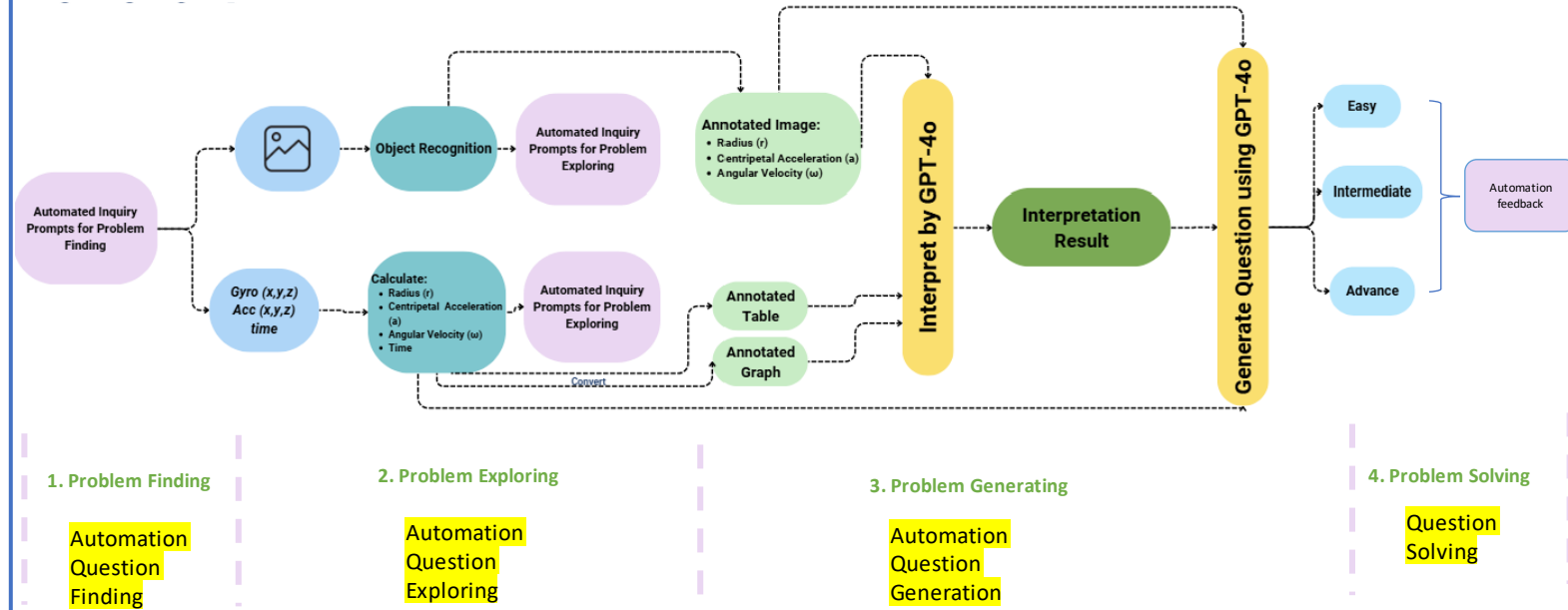
Center of the Framework: Sustainability & Scalability

Sustainability & Scalability are the core concepts of this framework. By enabling students to access context-based physics learning beyond the classroom and continuously track their progress, this system not only provides **more sustainable education** but can also be **scaled** to be used across different contexts and student populations.

P-MAGIC V.1



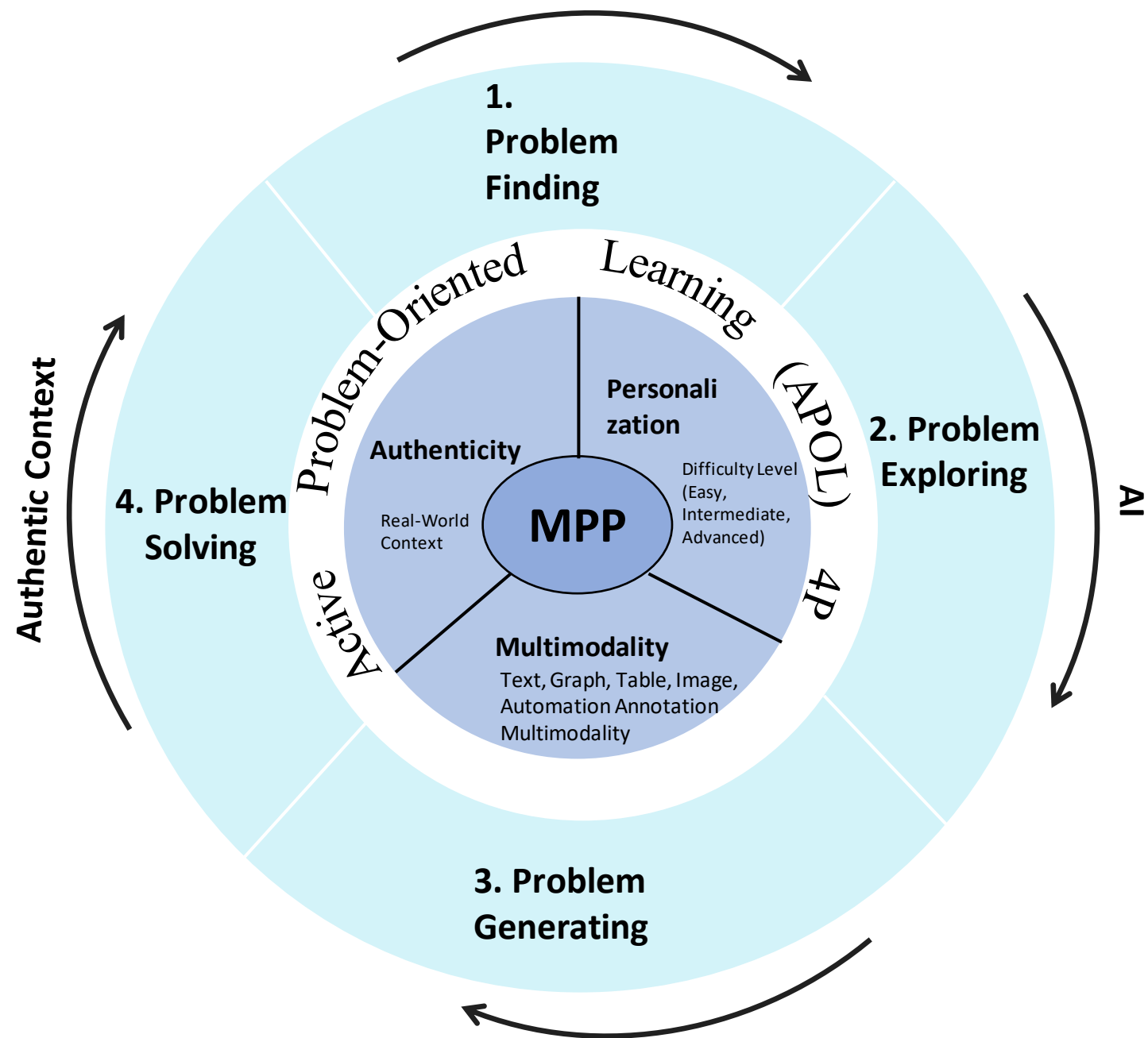
P-MAGIC V.2



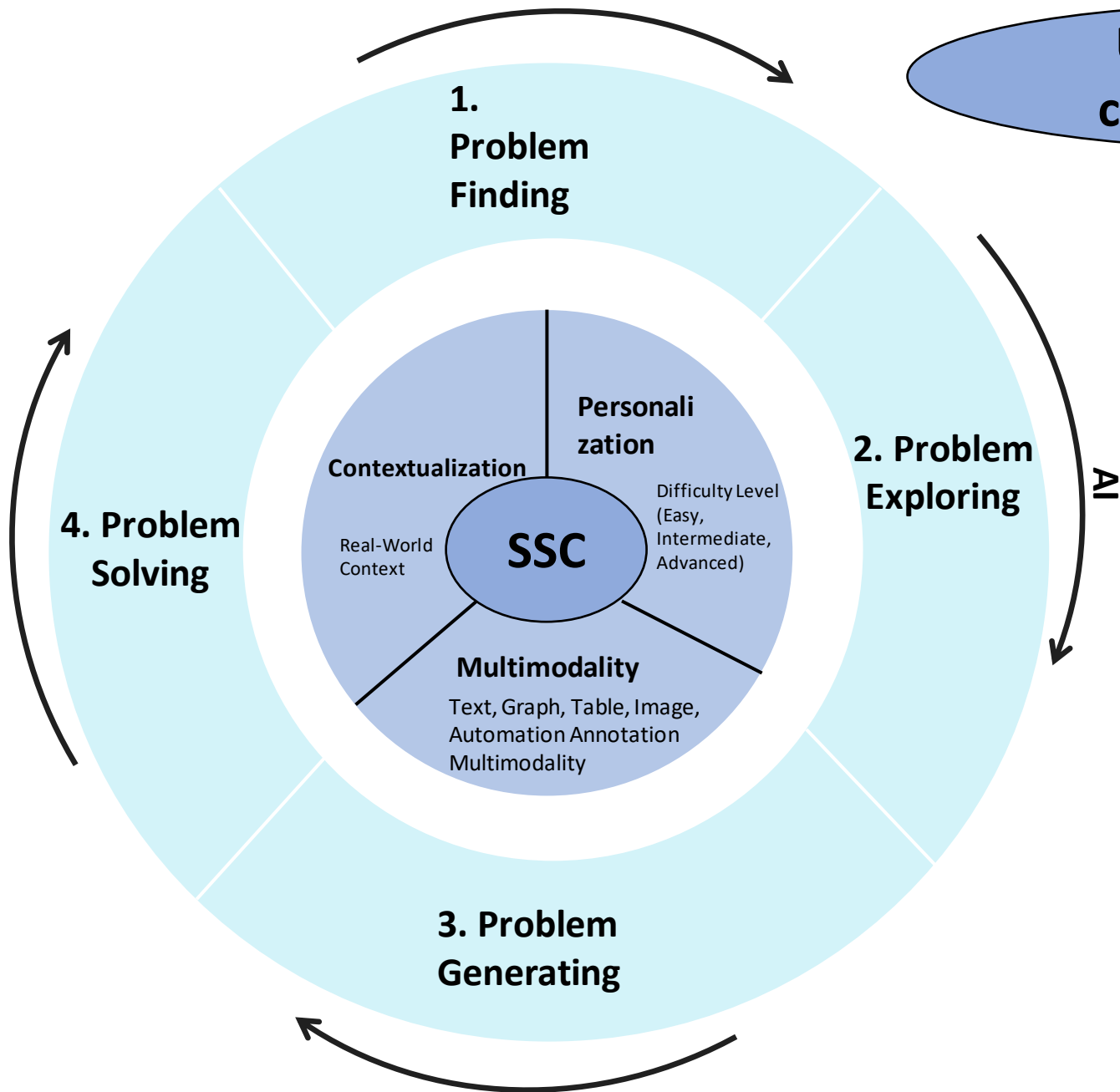
Modifications Needed:

1. Add Inquiry Prompts for Problem Finding, and Problem Exploring to stimulate active exploration and investigation.
2. Create a mechanism where we can guide learners in identifying physical phenomena around them that relate to centripetal acceleration (for problem exploration).
3. Add automation feedback in problem solving phase

Need revised



Authentic Context



Option names :

1. Finding–Exploration–Generation–Solving (FEGS) Cycle
2. Four-Stage Problem Cycle: Finding, Exploring, Generating, Solving
3. Real-World Multimodal Problem Workflow: Find, Explore, Generate, Solve
4. Authentic Multimodal Physics Problem Cycle (AMPPC)
5. Active Problem-Oriented Learning (APOL)
6. AI-Supported Authentic Physics Problem Cycle
7. Problem Discovery-to-Solution Cycle

Plan to do :

- 1. implementation automation problem finding feature in P-Magic system V.2
- 2. implementation automation problem exploring feature in P-Magic system V.2
- 3. implementation automation feedback in problem solving feature in P-Magic system V.2
- CONSIDER ONE MORE PHYSICS CONCEPT

Plan Quasai-Experiment :

For Senior High School Grade 11 students in Indonesia, the centripetal acceleration topic typically falls under the physics curriculum in the **fall semester (July-August)**. The topic would cover concepts such as the definition of centripetal acceleration, its relation to circular motion, and practical applications in real-life scenarios. It would be a great opportunity to incorporate P-Magic V.2

11-12-2025 Progress

- Inquiry-based learning
- Proposed Framework for Sustainable Problem Finding and Solving with AI

Inquiry-based theory is a way of learning where students discover ideas by **asking questions**, not just memorizing answers. Instead of being told how something works, students explore by asking “why,” “how,” and “what if.” These questions guide them to investigate, test ideas, think critically, and build their own understanding of a science concept.

In physics, inquiry-based questions help students look deeper into everyday phenomena.

By asking and exploring these questions, students learn physics through curiosity, investigation, and evidence—just like real scientists. [Howes et al. \(2008\)](#)

Authentic Contextual Exploration-based (need to check carefully) we use inquiry approach to asking (why, how)

Need to connect some concept formula to authentic context, (different with inquiry-based),
Automatically QnA, related to their data

Benefits of Inquiry-Based Learning [\(Qablan et al., 2024\)](#)

Encourages engaged learning

Students become more interested and active, which helps them gain knowledge, skills, and positive learning attitudes.

Fosters critical thinking

Students investigate questions, strengthening their critical-thinking abilities.

Sparks creativity

Students have freedom to explore their own interests, often creating new and original solutions—especially in open-ended tasks.

Enhances problem-solving skills

Students learn to handle real-world problems by thinking creatively and developing innovative solutions.

Connects learning to real life

Students explore issues around them, helping them see how classroom concepts apply to real-world situations.

Framework for Sustainable Problem Finding and Solving with AI

This demonstrates our unique contributions to achieving sustainability and scalability through problem exploration, finding, generation, and solving in authentic contexts.

1. Problem Finding

Students are encouraged to **find physics concepts** in their surrounding environment related to centripetal acceleration. Using P-MAGIC, they can observe centripetal principles. This leads to **increased active engagement** and **motivation** to continue learning. This phase leverages **P-MAGIC models** to detect **misconceptions** and assist students in finding relevant and interesting physics problems.

2. Problem Exploring

In this stage, students are encouraged to **explore their choice of problems in problem finding stage**. Using P-MAGIC, they can experience and analyze the real world data centripetal principles through hands-on experiments. This leads to **increased active engagement** and **motivation** to continue learning.

3. Problem Generating

Next, **P-MAGIC generates problems** related to the object already find and explored. **Multimodal problems** including annotation images, text, graphs, or table, all aimed at testing a deeper understanding of the concepts. Include : **Problem difficulty level**.

4. Problem Solving

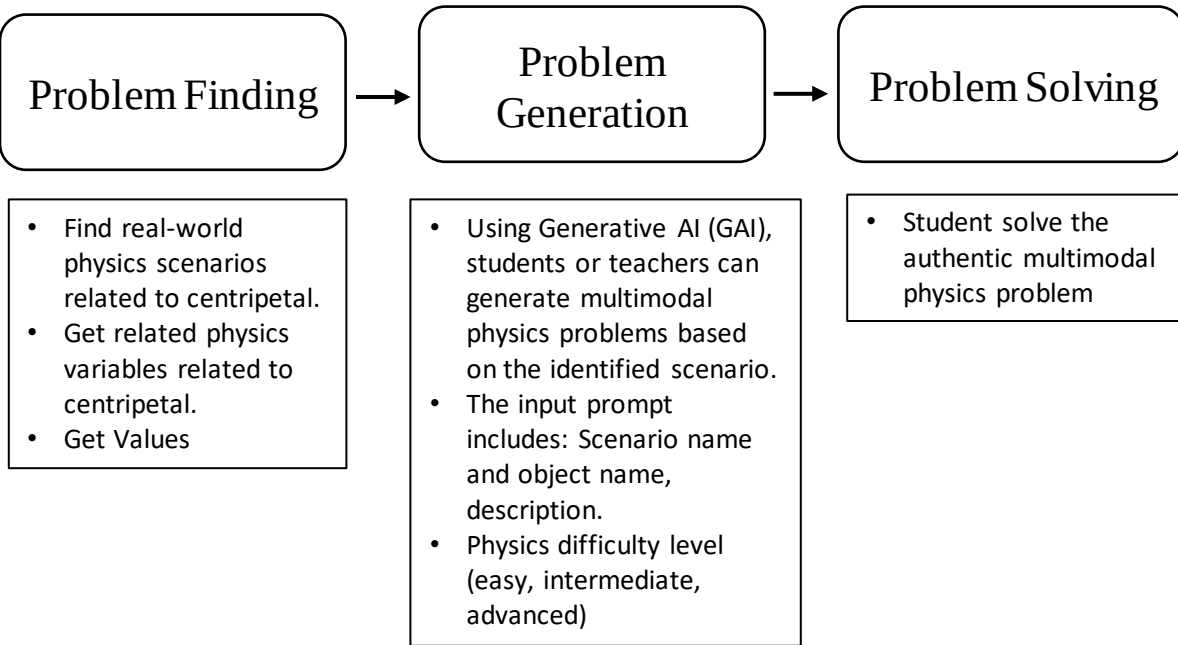
In this phase, students **solve the problems** generated by the system, receiving adaptive feedback. The **adaptive feedback system** offers clarifications and motivation when students encounter difficulties or show declining engagement. **Solutions based on real-world data** ensure **sustained learning**, connecting physics theory to real-world applications.

Center of the Framework: Sustainability & Scalability

Sustainability & Scalability are the core concepts of this framework. By enabling students to access context-based physics learning beyond the classroom and continuously track their progress, this system not only provides **more sustainable education** but can also be **scaled** to be used across different contexts and student populations.

2nd paper Framework

Active Problem-Oriented Learning (APOL)



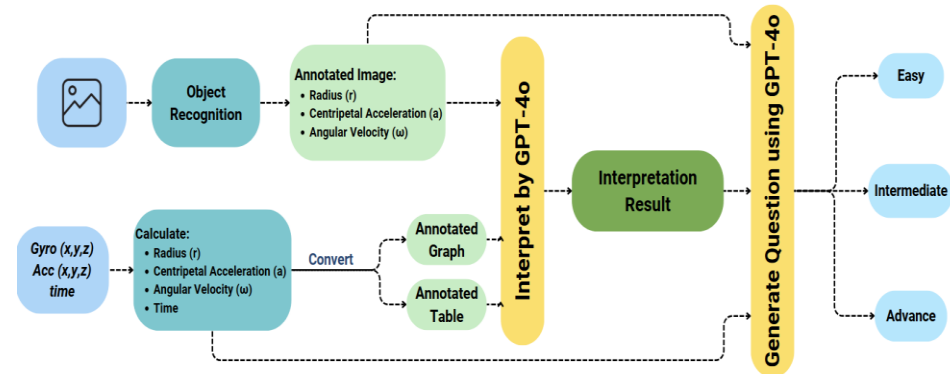
This study introduces P-MAGIC (Physics Multimodal Authentic Generated Interactive Content Problem), an AI-driven multimodal system designed to support authentic and personalized physics learning based on the Active Problem-Oriented Learning (APOL) framework.

3rd paper Framework

Active Problem-Oriented Learning (APOL) 4P

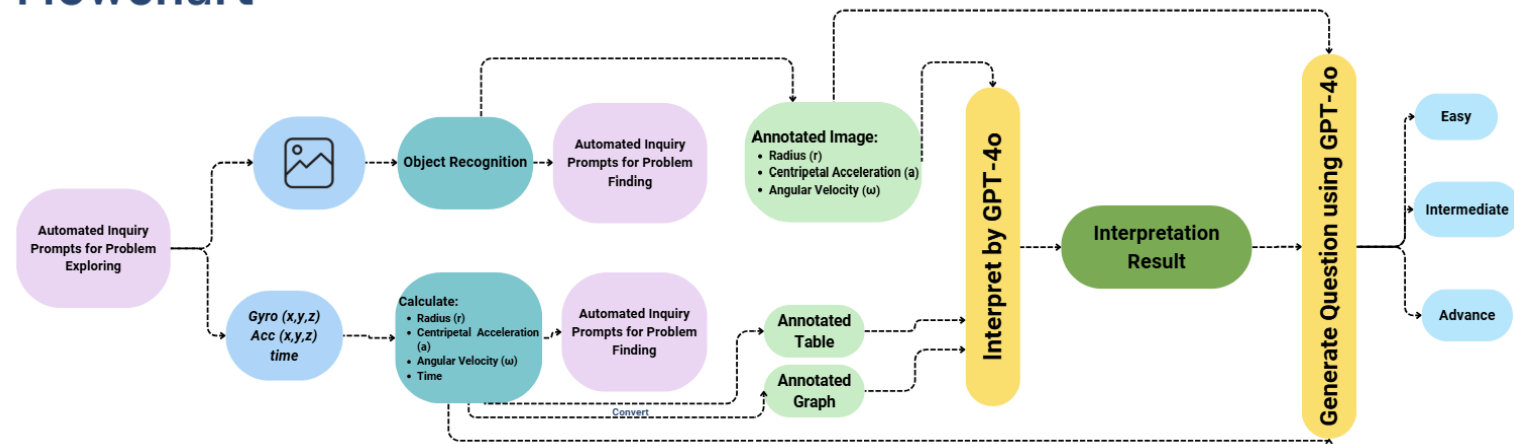
Need revised

P-MAGIC V.1



P-MAGIC V.2

Flowchart



Modifications Needed:

1.Add Inquiry Prompts for Problem Exploring and Problem Finding to stimulate active exploration and investigation.

2.Create a mechanism where we can guide learners in identifying physical phenomena around them that relate to centripetal acceleration (for problem exploration).

Problem Exploring:

Inquiry Prompt: The system can ask the learner to explore their surroundings for any object or phenomena that may involve rotational motion. For instance, asking students if they notice any rotating objects (e.g., fans, wheels, or playground equipment) that are related to centripetal acceleration.

Example Prompt for Inquiry:

- “Is there an object or situation in your surroundings that you think might be related to rotational motion or centripetal acceleration?”
“What do you observe in their motion?”

Problem Finding:

Inquiry Prompt: Once the student has explored and noticed a real-world object, the system will ask them to **find the physics problem** related to the observed phenomena.

Example Inquiry:

"Based on what you “observed” (automation recognition object name), how would you describe the motion of the object?"

“Can you think of any physical quantities like angular velocity or radius that might describe the motion?”

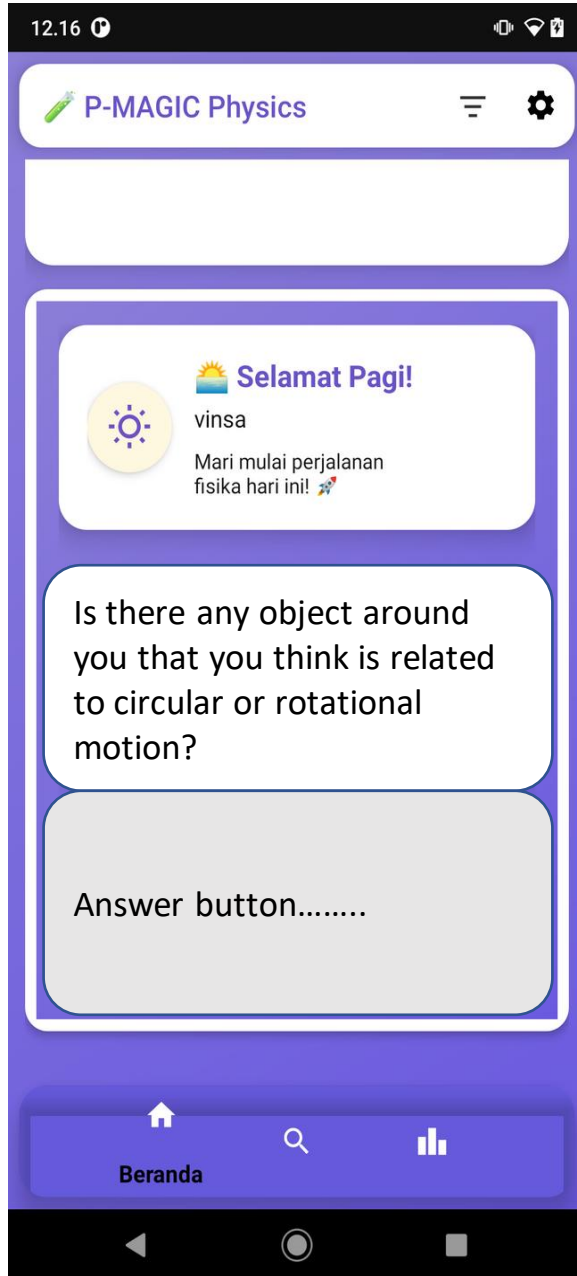
“How would you relate them to centripetal acceleration?”

P-MAGIC V.2

Automated Inquiry Question during Problem Finding Phase

P-MAGIC V.2

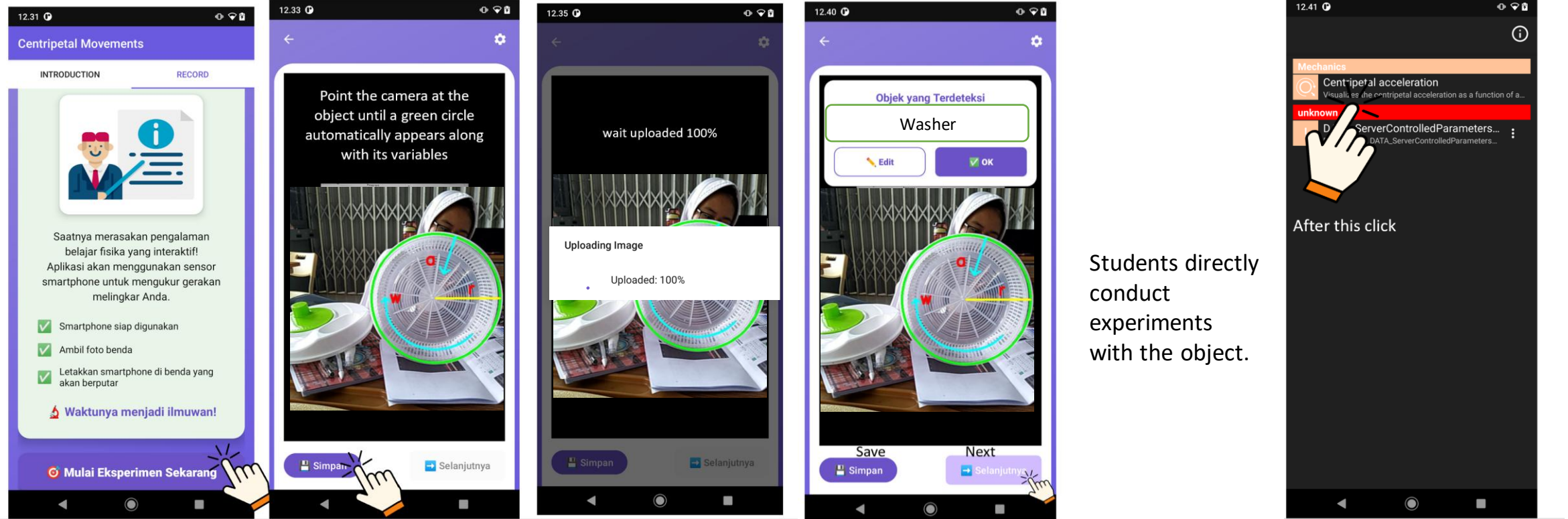
- Will add Automated Exploratory Inquiry Question



P-MAGIC V.2

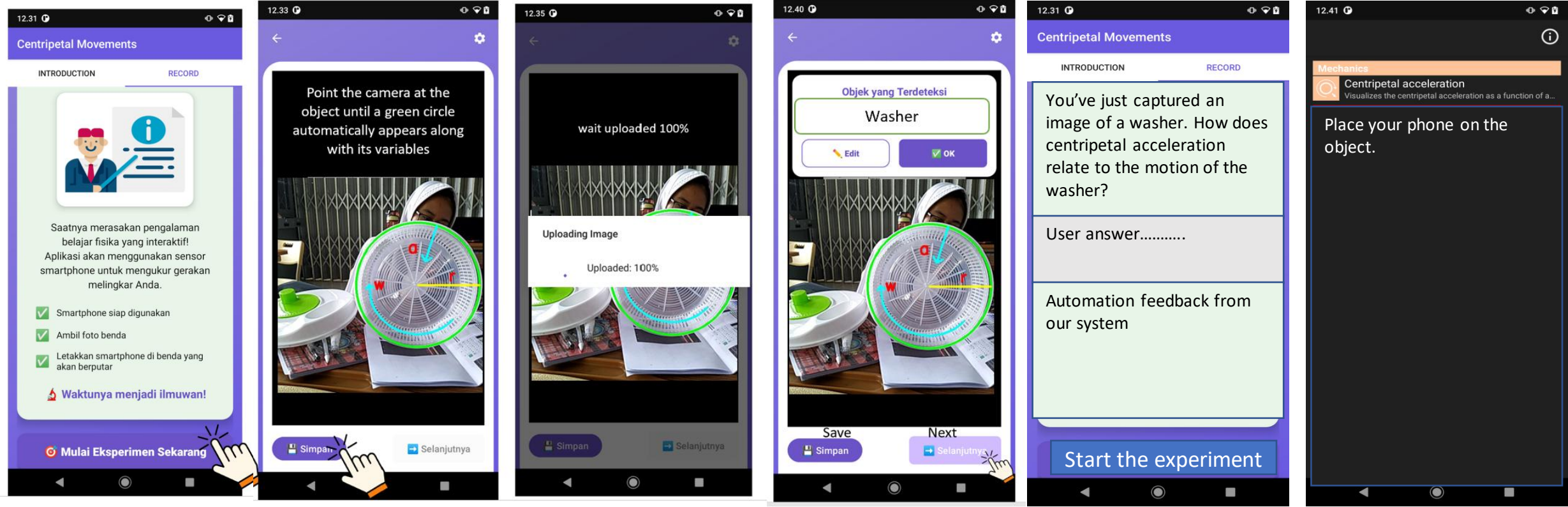
Automated Inquiry Question during Problem Exploring Phase

- **Without** Automated Exploratory Inquiry Question



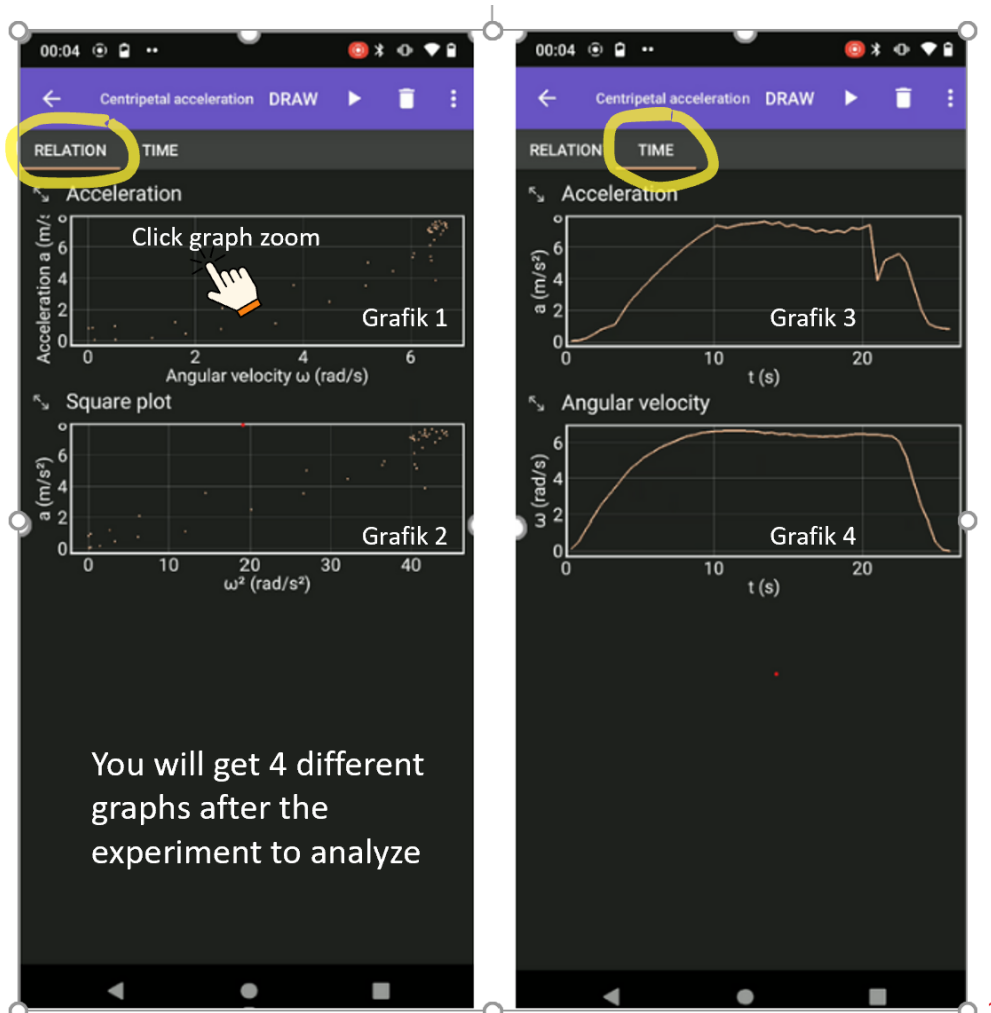
P-MAGIC V.2

- Will add Automated Exploratory Inquiry Question and automatically feedback user's answer



P-MAGIC V.1

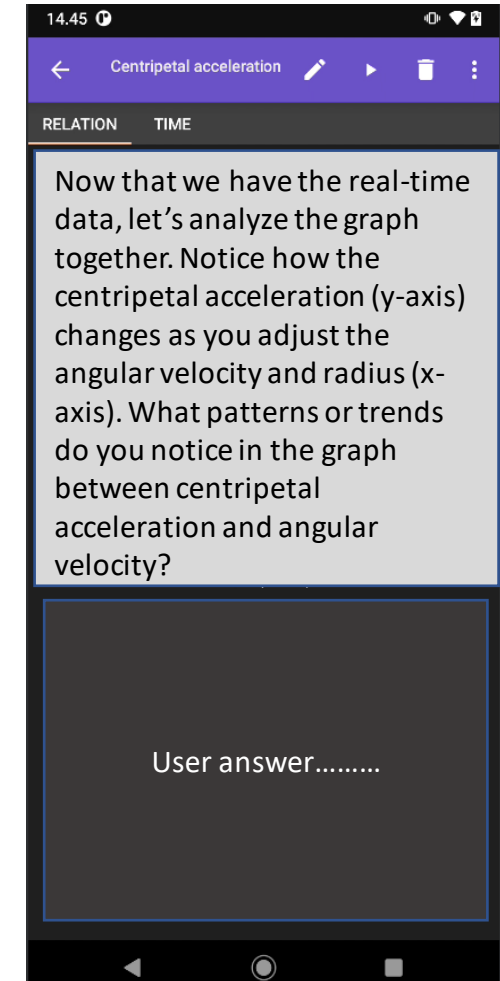
- **Without** Automated Exploratory Inquiry Question



Once the graph is generated from the real-world experiment, the system automatically provides students with the MPP.

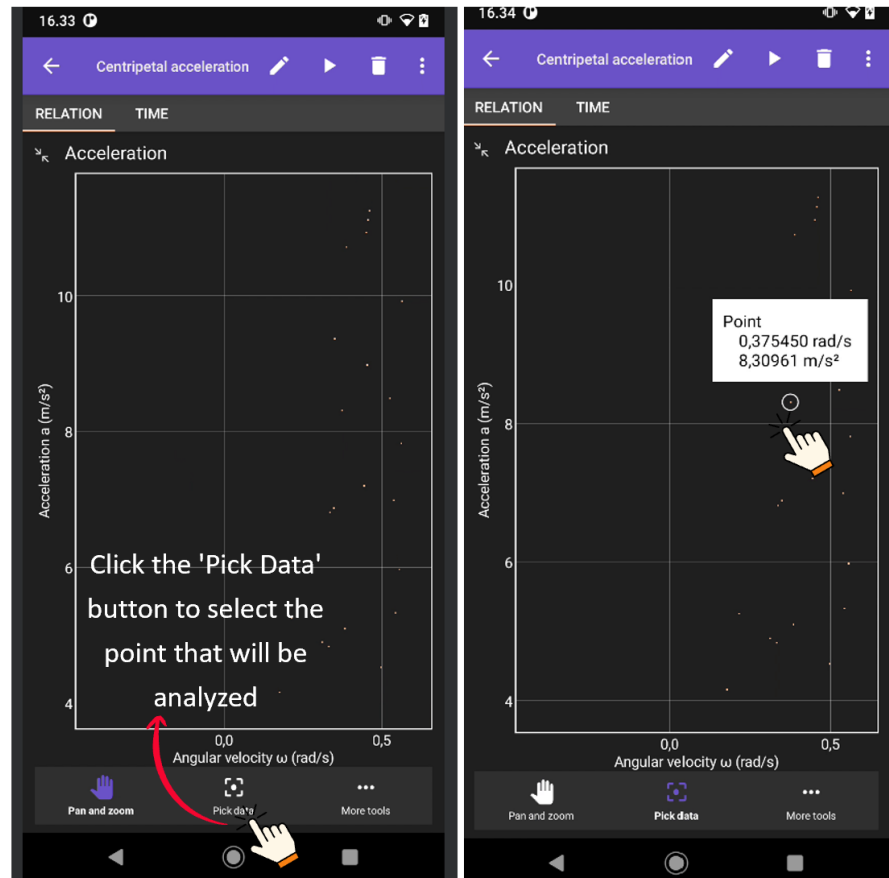
P-MAGIC V.2

- Will add Automated Exploratory Inquiry Question

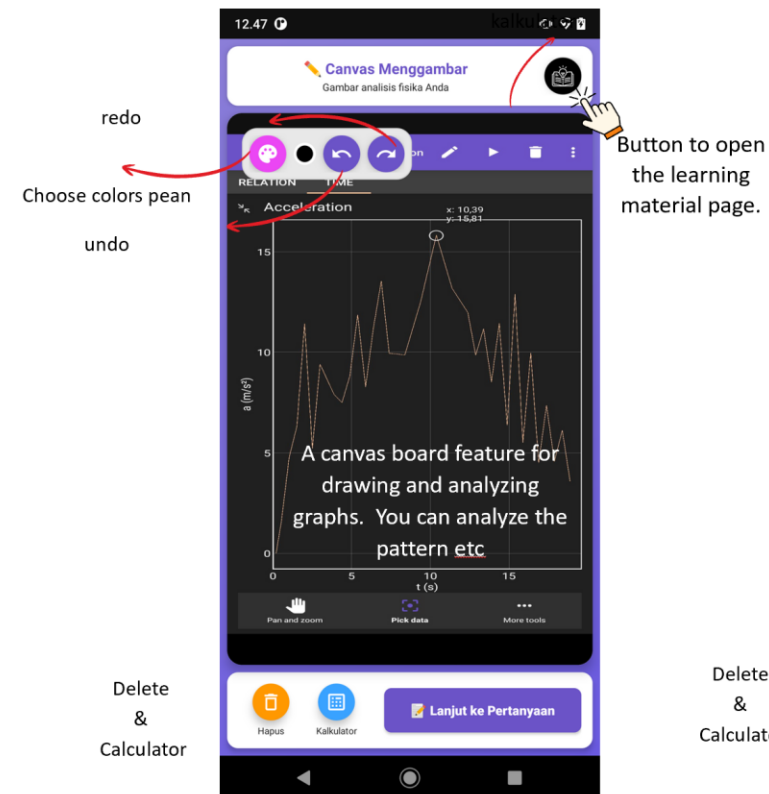


- Without Automated Exploratory Inquiry Question**

The user is asked to choose from four graphs representing the relationships between centripetal variables and select the data values to analyze further



The user is asked to analyze the graph and the values they have selected in order to analyze and calculate the radius from the given data.



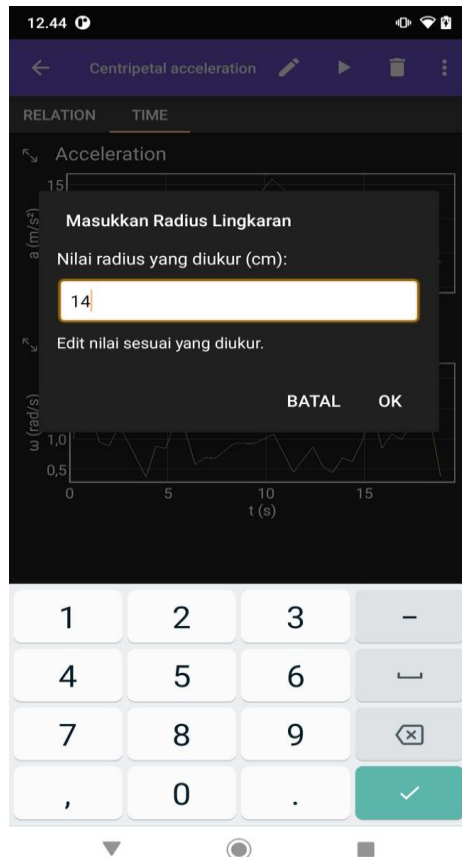
A canvas board feature for drawing and analyzing graphs. Calculate the radius if the values inside the x, y points on the graph are known

If done, click next

P-MAGIC V.1

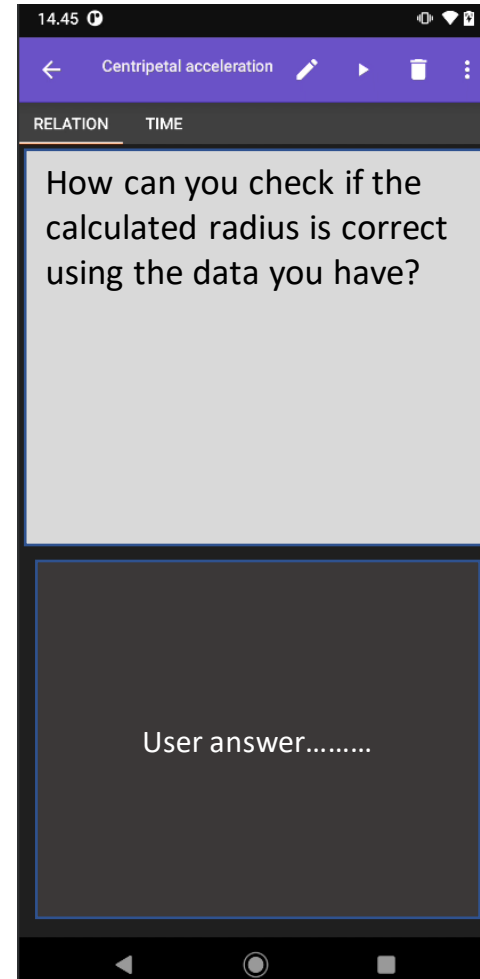
- **Without** Automated Exploratory Inquiry Question

The radius of the object will be calculated automatically. However, if it is inaccurate, you can edit it to match the measured radius of the object



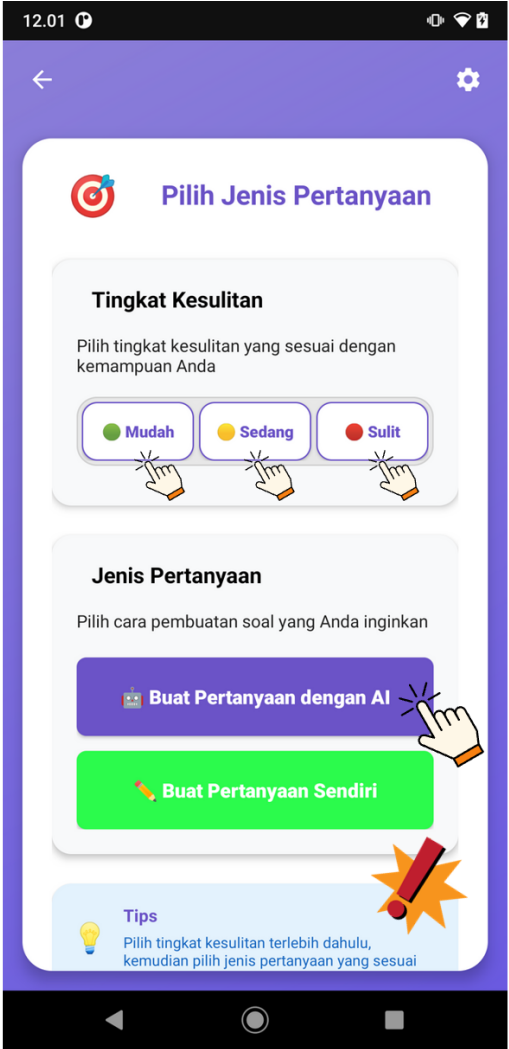
P-MAGIC V.2

- Will add Automated Exploratory Inquiry Question

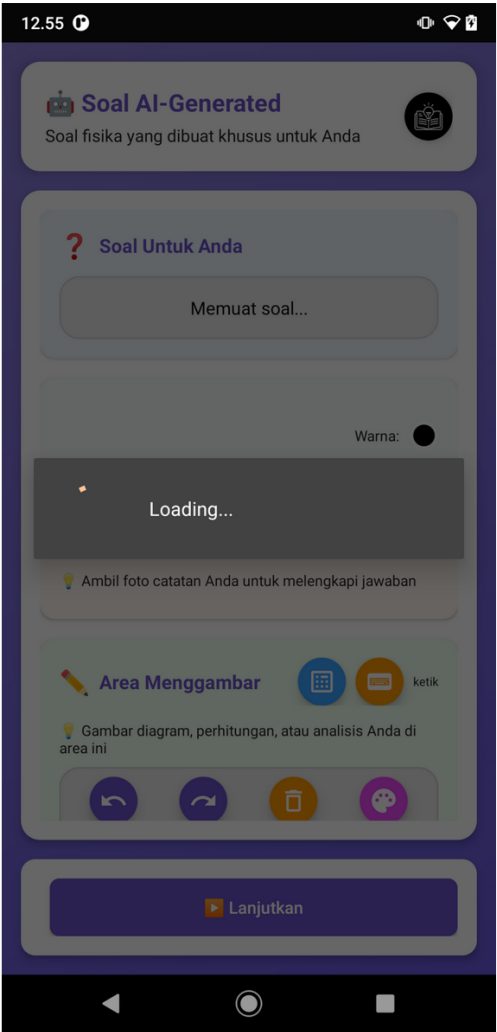


Problem Generation Phase

- Choose the difficulty Level

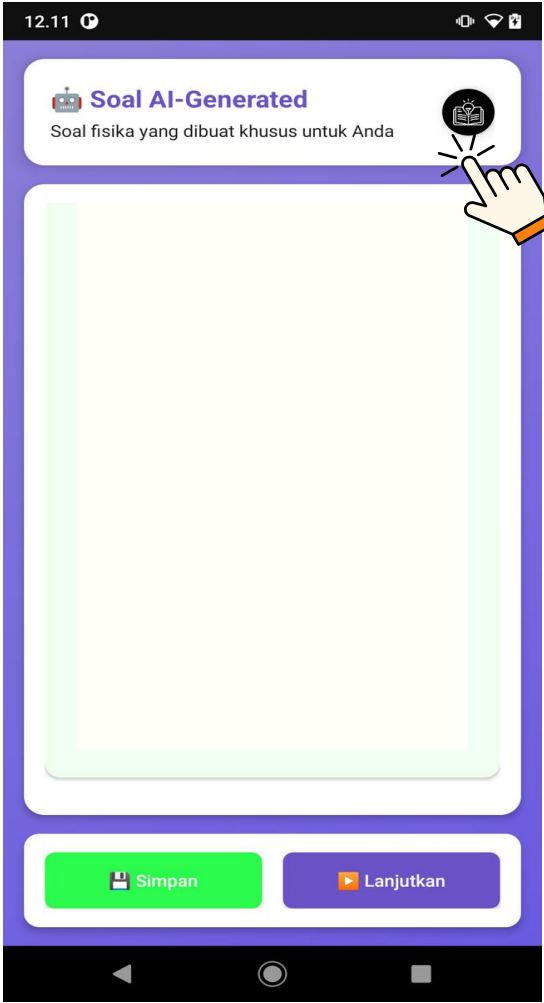
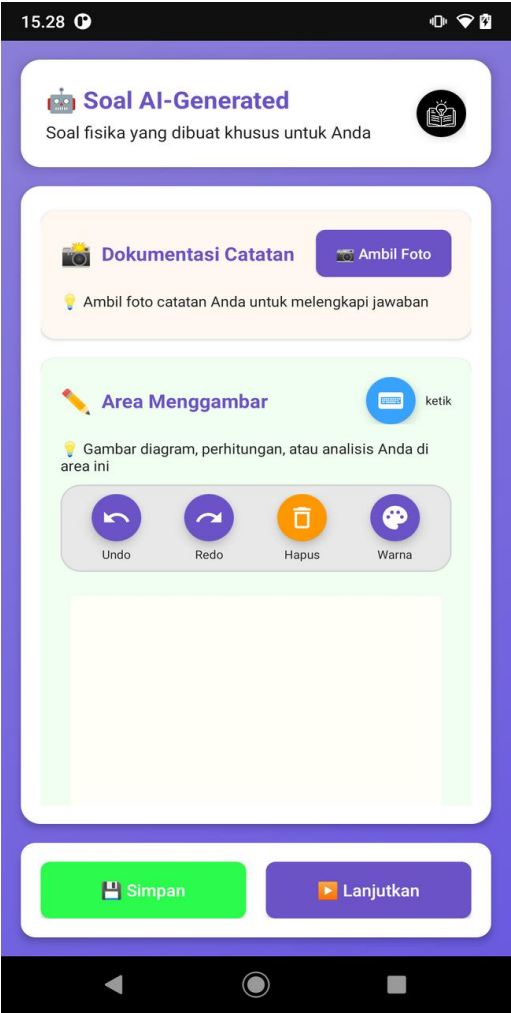


- Multimodal Physics Problem Generation



Problem Solving Phase
+ Giving automatic feedback

Answer page: The Canvas and typing button menu are available.



Button to open the learning material page.

